

# Sandipan Seal

AI / Machine Learning Engineer — LLM & Agentic Systems

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## Summary

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AI/ML engineer with 3+ years building **production LLM and agentic systems** — RAG pipelines, multi-agent applications, and safety-guarded LLM gateways deployed on AWS and Kubernetes. My master's research drove LLM-based reasoning to **99.4% jamming-regime classification accuracy**, beating classical ML baselines by 20+ points while keeping every explanation fully grounded — evidence I can turn frontier AI into reliable, measurable systems. Now focused on **Forward-Deployed / AI Engineer** roles, partnering directly with customers to take ambiguous GenAI use cases from prototype to production.

## Skills

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**Languages:** Python, TypeScript, JavaScript, Go, C++, SQL

**LLM / GenAI:** RAG, multi-agent & agentic systems, prompt engineering, tool use / function calling, human-in-the-loop, fine-tuning, embeddings & vector search, evaluation (LLM-as-judge, RAGAS), guardrails (PII redaction, prompt-injection defense), AI risk & data privacy (GDPR), LLMops

**LLM Frameworks & Providers:** LangChain, CrewAI, Hugging Face, sentence-transformers, OpenAI (GPT-4o), Ollama / vLLM

**ML / Data Science:** PyTorch, TensorFlow, scikit-learn, pandas, NumPy, computer vision, NLP, CNNs, clustering (PCA/UMAP)

**Backend & Web:** FastAPI, Pydantic, SQLAlchemy, Node.js, React.js, Next.js, Tailwind CSS, REST APIs, microservices

**Databases:** PostgreSQL, Redis, Qdrant (vector DB), Neo4j, MongoDB, Supabase

**Cloud, DevOps & Observability:** AWS (Lambda, API Gateway, CloudFront, S3, SQS), Terraform (IaC), Docker, Kubernetes, CI/CD (GitHub Actions), Prometheus, Grafana, OpenTelemetry, Langfuse, SonarQube

**Workflow & Testing:** n8n, Playwright, pytest, Locust, API testing

## Experience

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**Working Student — AI/ML Software Development, aiiio GmbH** — Magdeburg, Germany *Jan 2025 – Present*

- Engineered **LLM-powered, RAG-enabled multi-agent systems** (Python, FastAPI) that **automated the test-generation lifecycle**, cutting manual test-authoring effort ~40% by converting natural-language requirements into executable test suites.
- Accelerated delivery of customer-facing AI tools by building responsive **React.js / Next.js** front-ends integrated with production models.
- Automated company segmentation across **1,000+ companies** by building a clustering pipeline (**TF-IDF, multilingual MiniLM-BERT embeddings**, PCA/UMAP, K-means/DBSCAN) validated with silhouette/purity metrics and German WZ2008 sector reporting.

**Senior Software Engineer, Persistent Systems Ltd** — Kolkata, India

*Sep 2021 – Mar 2024*

- Raised healthcare-diagnostic prediction accuracy ~20% for client Adaptive Biotechnologies by designing and **deploying predictive ML models** into production.
- Delivered end-to-end **full-stack solutions** for enterprise clients (PerkinElmer) across **React, .NET/C#**, and **MongoDB**, owning features from API design to production deployment.
- Cut regression-test cycle time ~30% by building automated **QA** suites (Selenium, API testing); twice awarded the **High Five Team Award**.

## Selected Projects

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**InferOps-AI — Production LLM Gateway** · github.com/sandipanseal/InferOps-AI

- Cut LLM inference cost and blocked unsafe requests via **cost-aware model routing**, **RAG** (Qdrant/pgvector), **PII redaction**, and **prompt-injection guardrails**, verified by a deterministic + **LLM-as-judge** + **RAGAS** eval suite at **>80% test coverage**.
- Shipped to production on AWS serverless (**Lambda, API Gateway, CloudFront, S3**) via **Terraform**, with **Langfuse** tracing and Prometheus/Grafana observability; FastAPI, **LangChain** (ReAct), PostgreSQL, Redis, Qdrant, Ollama/OpenAI/vLLM, Docker/Kubernetes.

**AegisOps-AI — Agentic Incident Commander** · github.com/sandipanseal/AegisOps-AI

- Reduced incident-triage effort** by building a **multi-agent** system that performs automated **root-cause analysis** over logs, metrics, and **Kubernetes** state, with confidence scoring and **approval-gated runbooks** (**human-in-the-loop**).

- Implemented SLA tracking, a lifecycle state machine, and service-dependency / blast-radius analysis; FastAPI, Next.js 15, OpenTelemetry, pytest + Playwright e2e.

**Master's Thesis — Explainable Jamming Analysis via LLM-Based Evidence Interpretation · OVGU Magdeburg**

- **Achieved 99.4% regime and 99.9% binary jamming-detection accuracy** on a balanced held-out set — **outperforming Logistic Regression (79%) and Random Forest (65%)** — by using an **LLM as an evidence interpreter** over engineered **RF embeddings**, not a raw classifier.
- **Guaranteed 100% valid JSON output and 100% evidence-grounding** by enforcing a strict output contract and validating it with grounding and perturbation-based faithfulness tests.

**AI Automation Testing Tool (CrewAI) — Multi-Agent QA · [github.com/sandipanseal/AI-Automation-Testing-Tool-using-CrewAi](https://github.com/sandipanseal/AI-Automation-Testing-Tool-using-CrewAi)**

- **Cut manual test creation** by building a **CrewAI multi-agent** system that converts natural-language requirements into executable **Playwright** tests (explorer, scenario writer, script generator, auto-repairing runner) with live SSE streaming.

**Additional projects:** AI Compliance Gatekeeper (**n8n** + FastAPI — PII/policy classification & **data-leakage prevention**) · Mine@TENJI (German parliamentary speech recommender — **SentenceTransformers**, **FAISS**). All on GitHub.

## Education

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**M.Sc. Digital Engineering (Computer Science), Otto-von-Guericke-University (OVGU) Magdeburg — Germany** *Apr 2024 – Oct 2026 (expected)*

**B.Tech. Electronics & Communication Engineering, Narula Institute of Technology — Kolkata, India** *Aug 2017 – Jun 2021*

## Awards & Patent

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**Patent:** Low-Cost Automated Bag-Valve-Mask Ventilation Machine — Indian Patent IN 202031037602 A (2020)

**Awards:** Conquer COVID-19 Hackathon Winner (2020) · Best Paper ×2 (I3SET 2019, NCMR 2020) · High Five Team Award ×2 (Persistent)

## Certifications

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Claude Code in Action — Anthropic (2026) · Fine-tuning LLMs — Hugging Face (2026) · AI Agent Fundamentals — Databricks (2026) · Certified Ethical Hacker (CEH) — EC-Council (2020)

## Languages

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Bengali (Native, C2) · English (IELTS, C1) · German (A2)